DUAL-CORE OPTICAL MOTION TESTBED

FIRMWARE, HOST SOFTWARE & RESULTS

STM32H747 MotionIPC, live GUI, telemetry, reliability, and characterization

Customer-facing technical report
Revision 2.9 - 4 August 2026

Prepared by	Herder Elektronische Systemen
Primary subject	Dual-core firmware, host GUI, telemetry, reliability controls, and measured operation
Evidence basis	MotionIPC source, motion firmware, GUI video, plots, logs, slide reviews, and reports
Distribution	Customer-facing; selected source excerpts and engineering evidence included

DOCUMENT PURPOSE

This report presents the software-controlled Dual-Core Optical Motion Testbed: dual-core firmware, target-side motion control, live operator software, data integrity, safety behavior, and measured results. Hardware and PCB implementation are documented only in the companion platform report.

1. System Outcome

The completed system combines a custom STM32H747 dual-core IPC architecture with target-side motion control, encoder acquisition, host supervision, live visualization, structured logging, automated workflows, and postprocessing. The 400 MHz Cortex-M7 and 200 MHz Cortex-M4 communicate through OpenAMP/RPMsg for coordination and a custom AXI-SRAM ring for motion records.

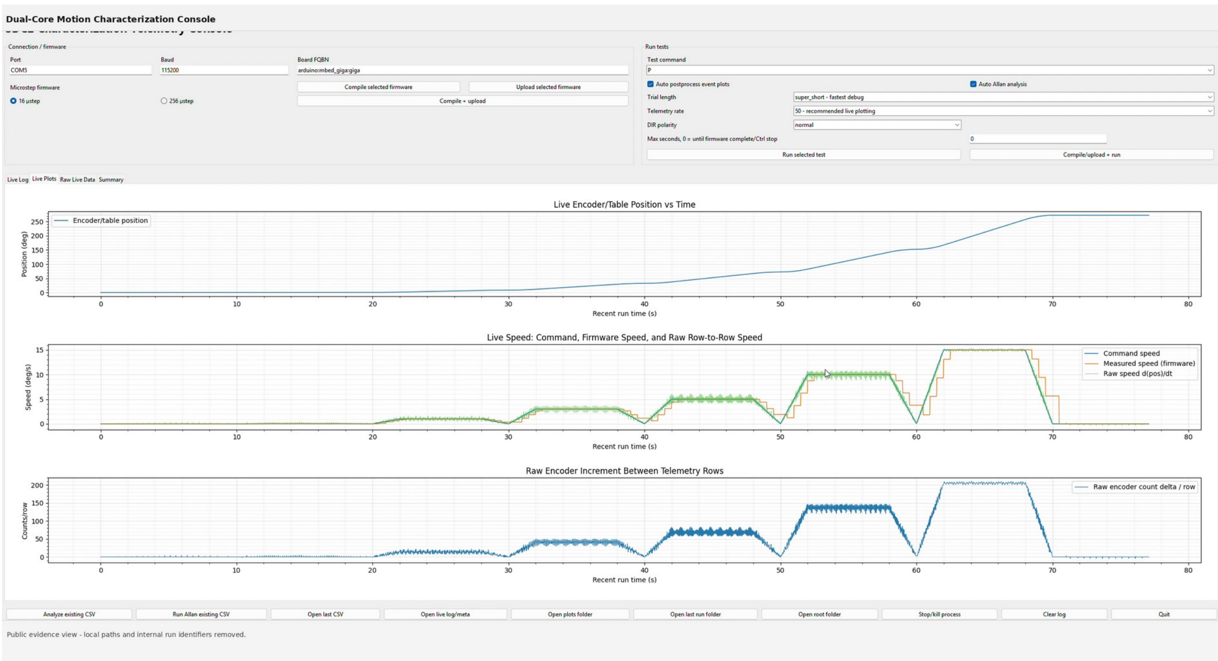

Figure 1. Operator-facing host application with connection controls, test selection, status, and live telemetry.

CUSTOMER-VISIBLE RESULT

An operator can connect the target, select a test, monitor command and encoder behavior, record timestamped data, recover interrupted runs, and launch analysis while real-time motion remains on the embedded target.

2. Dual-Core Architecture

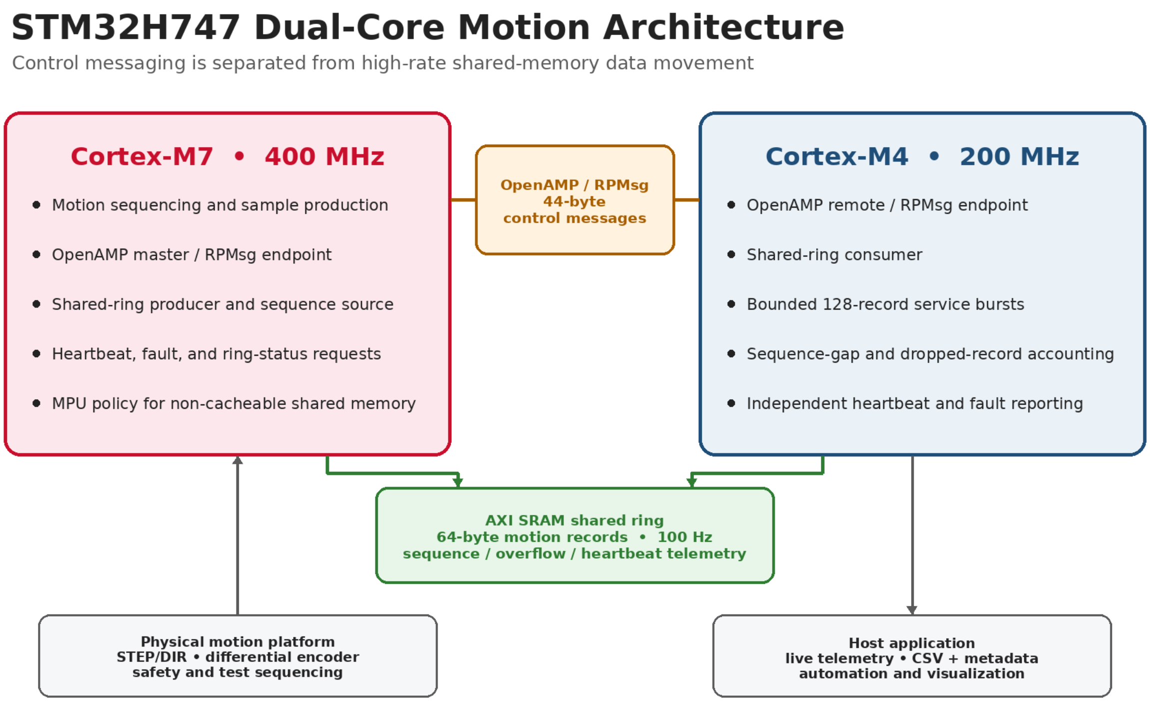

Figure 2. STM32H747 architecture separating low-rate OpenAMP/RPMsg coordination from the AXI-SRAM motion-data path.

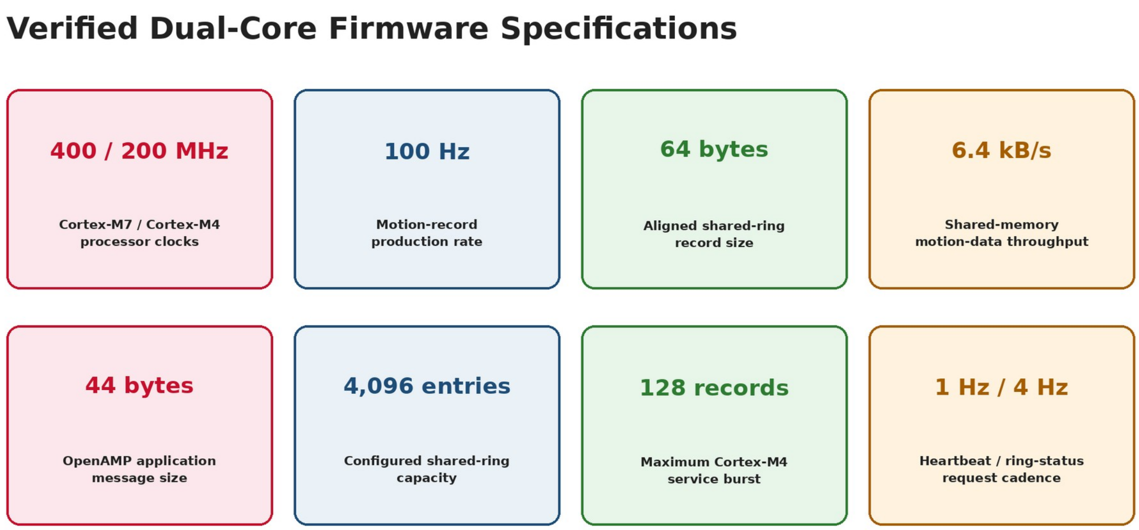

Figure 3. Source-verified processor, IPC, buffering, and service-cadence specifications.

3. Core Ownership, IPC, and Memory

Item	Verified implementation
Cores	400 MHz Cortex-M7 + 200 MHz Cortex-M4
Control IPC	OpenAMP/RPMsg; 44-byte application messages
High-rate IPC	256 KiB AXI-SRAM producer/consumer ring
Motion record	64 bytes, cache-line aligned
Record rate	100 Hz; 6.4 kB/s internal throughput
Consumer service	Up to 128 records per M4 service pass
Health cadence	1 Hz per-core heartbeat; 4 Hz ring-status requests

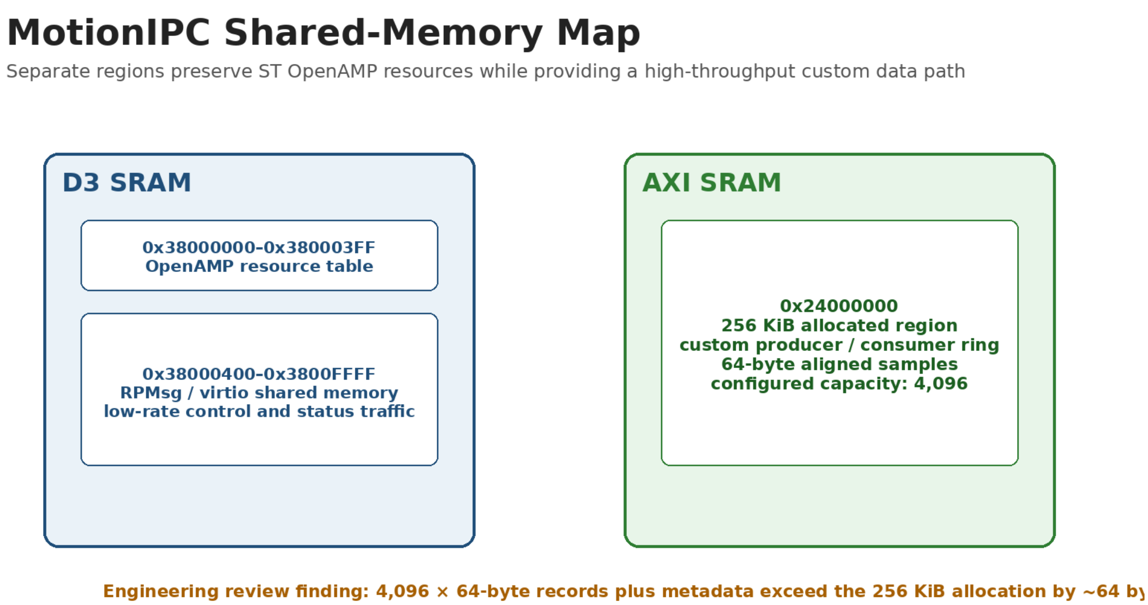

Figure 4. Shared-memory map and the identified ring-capacity correction.

ENGINEERING REVIEW FINDING

The configured 4,096 × 64-byte sample array fills the 256 KiB allocation before metadata is counted. A production release should use 4,095 entries or enlarge the reserved region.

4. Target I/O Boundary

The firmware owns the controller-facing signal boundary: interrupt-driven encoder inputs, target-generated STEP/DIR outputs, motion-state transitions, and fault-safe shutdown behavior. Electrical conditioning, connectorization, routed copper, and the dedicated ground plane are documented exclusively in the companion platform and electrical report.

Signal or boundary	Firmware responsibility
ENC_A / ENC_B	Interrupt-driven encoder capture and direction decoding
STEP / DIR	Target-generated motor command timing
5 V / GND reference	Assumed external logic interface boundary; electrical implementation in companion FRD
Emergency stop / deadman	Independent hardware stop plus target-side host-timeout response

SEPARATION OF CONCERNS

This report focuses on firmware behavior and software outcomes. PCB design evidence appears only in the Mechanical Platform & Interface PCB FRD.

5. Target Motion Firmware and Safety

Real-time pulse generation, encoder capture, test progression, and deadman behavior remain on the embedded target. The host requests actions and records results but does not schedule motion.

Control or safety item	Verified value
STEP pulse high time	4 µs
Step-rate safety cap	25,000 pulses/s
Maximum pulse backlog	25 pulses/pass
Loop-overrun warning	25 ms
Speed-estimation window	500 ms
Host heartbeat	500 ms
Target deadman timeout	2,000 ms
Position correction	20 pulse-Hz/degree; 250 Hz absolute cap; 5% relative cap

5.1 Encoder and drive scaling

Parameter	Value
Gear ratio	24.65:1
256-microstep pulses/table revolution	1,262,080
Command increment	~0.000285°/pulse
Encoder CPR	5,000
Effective table counts/revolution	246,500
Measured increment	~0.001460°/count

6. Host GUI and Live Operation

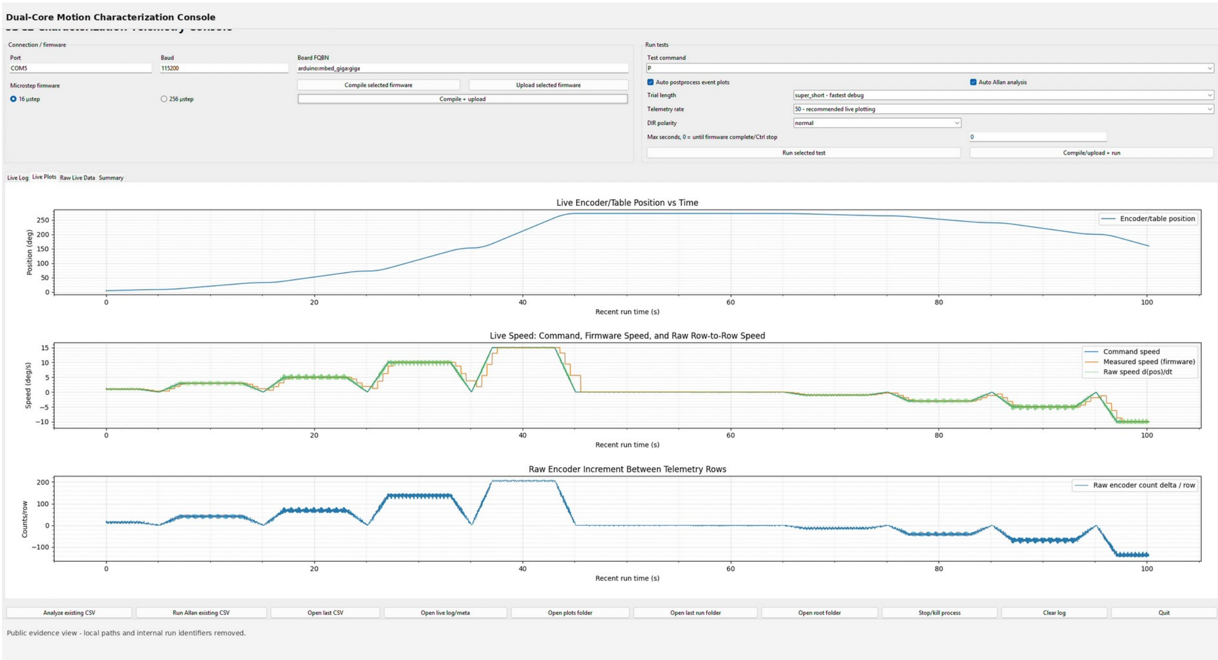

Figure 7. Active host-target test showing command, measured motion, encoder position/rate, status, and logging state.

The GUI integrates connection management, command execution, test selection, live plots, file naming, logging state, and operator feedback. It provides one working surface for the target instead of requiring separate serial terminals and analysis scripts.

- Live command-versus-measured speed and encoder-position plots.
- Raw encoder-rate visualization for diagnosing quantization and event timing.
- Target status, test progression, connection state, and completion markers.
- Automated filename generation, CSV capture, and separate metadata/comment retention.
- Supervision of tracking, static, step, Allan, and sine-response workflows.

7. Telemetry, Logging, and Recovery

Capability	Implementation
Host transport	115,200-baud serial
Selectable output	10, 20, 50, or 100 Hz requests
Record schema	24 timestamped CSV fields
Validation	Header and row-width checks; malformed-row detection
Run completion	Test-specific completion markers
Metadata	Controller comments and run context stored separately
Recovery	Partial-run preservation after serial interruption
Safety	500 ms heartbeat and 2 s target deadman

7.1 Bandwidth result

A 115,200-baud 8-N-1 link provides an idealized 11,520 payload bytes/s. Representative 24-field ASCII rows can approach 189 bytes, limiting sustained full-record output to roughly 50–60 rows/s. The 100 Hz selection therefore motivates shorter records, binary framing, or a faster external transport; the internal shared-memory path remains independent of this bottleneck.

8. Automated Test Workflow

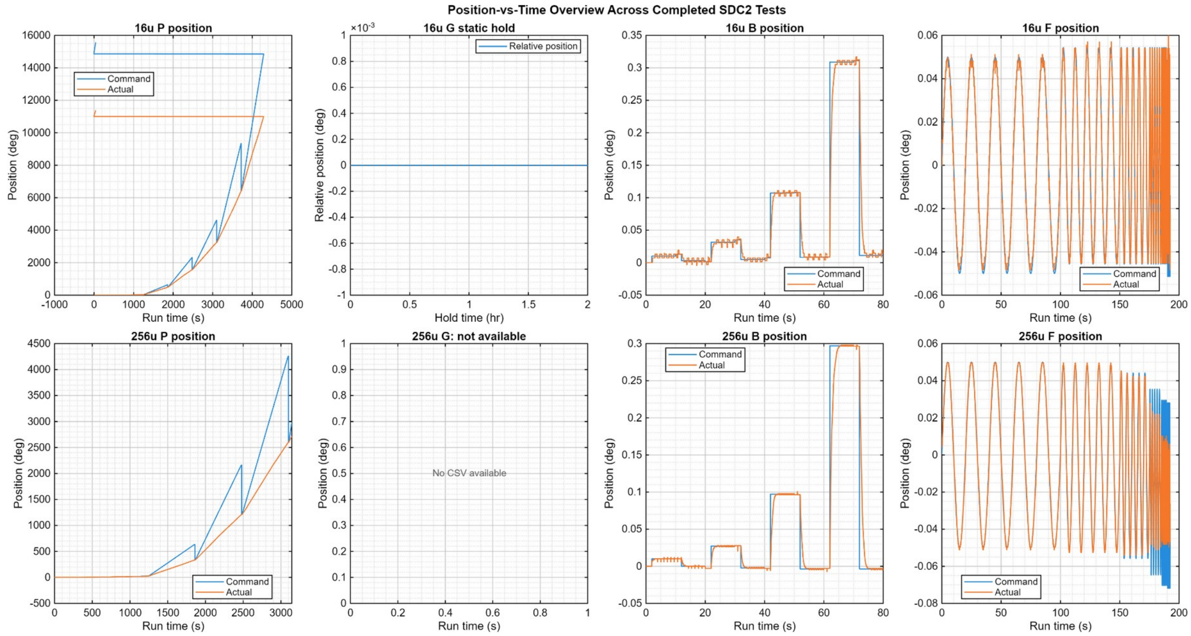

Figure 8. Representative scripted test sequences generated and recorded through the target/host workflow.

Workflow	Configured execution
Static hold	50 Hz × 180 s or 100 Hz × 90 s; 9,000 records
Dynamic tracking	100 Hz requested; 6,000 records per 60 s segment
Step response	Scripted steps with rise, overshoot, settling, RMS, and peak-error analysis
Allan analysis	Static position and dynamic residual workflows
Sine response	Closed-loop command/actual response over approximately 0.33–2 Hz

9. Low-Speed Tracking Results

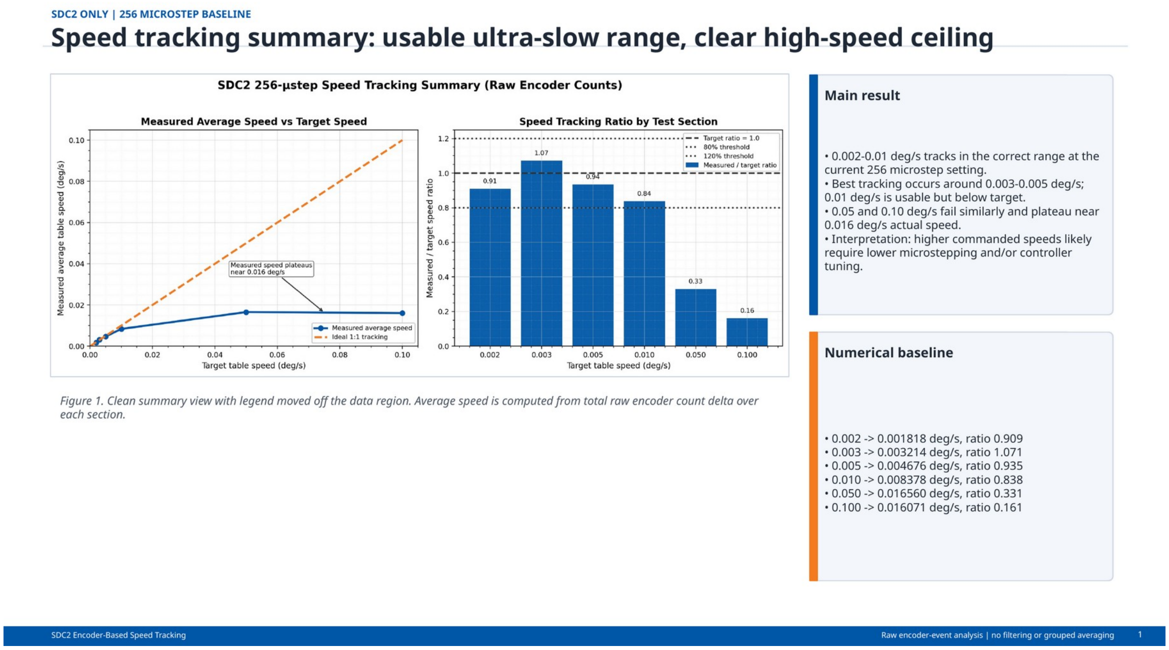

Figure 9. Measured low-speed performance in the 256-microstep configuration.

Command (deg/s)	Measured (deg/s)	Ratio / interpretation
0.002	0.001818	0.909
0.003	0.003214	1.071
0.005	0.004676	0.935
0.010	0.008378	0.838
0.050	~0.016	Configuration throughput ceiling
0.100	~0.016	Configuration throughput ceiling

RESULT

The 256-microstep setup is an ultra-slow configuration. Best tracking occurred near 0.003–0.005°/s; faster commands plateaued near 0.016°/s and require a different microstep or pulse-rate configuration.

10. Tracking, Step, and Frequency Results

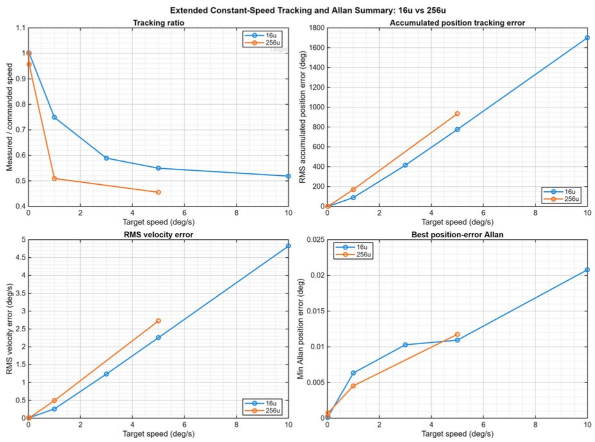

Figure 10. Constant-speed command and measured motion.

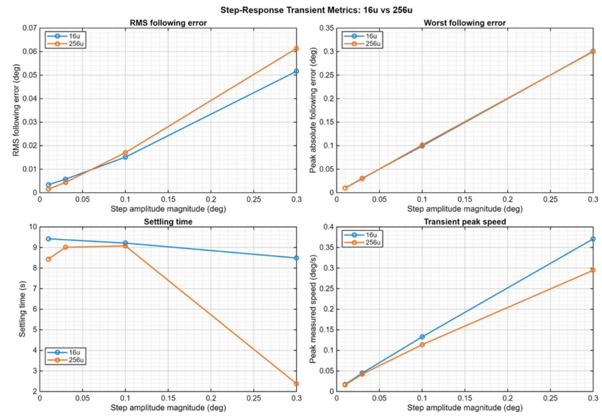

Figure 11. Scripted step-response characterization.

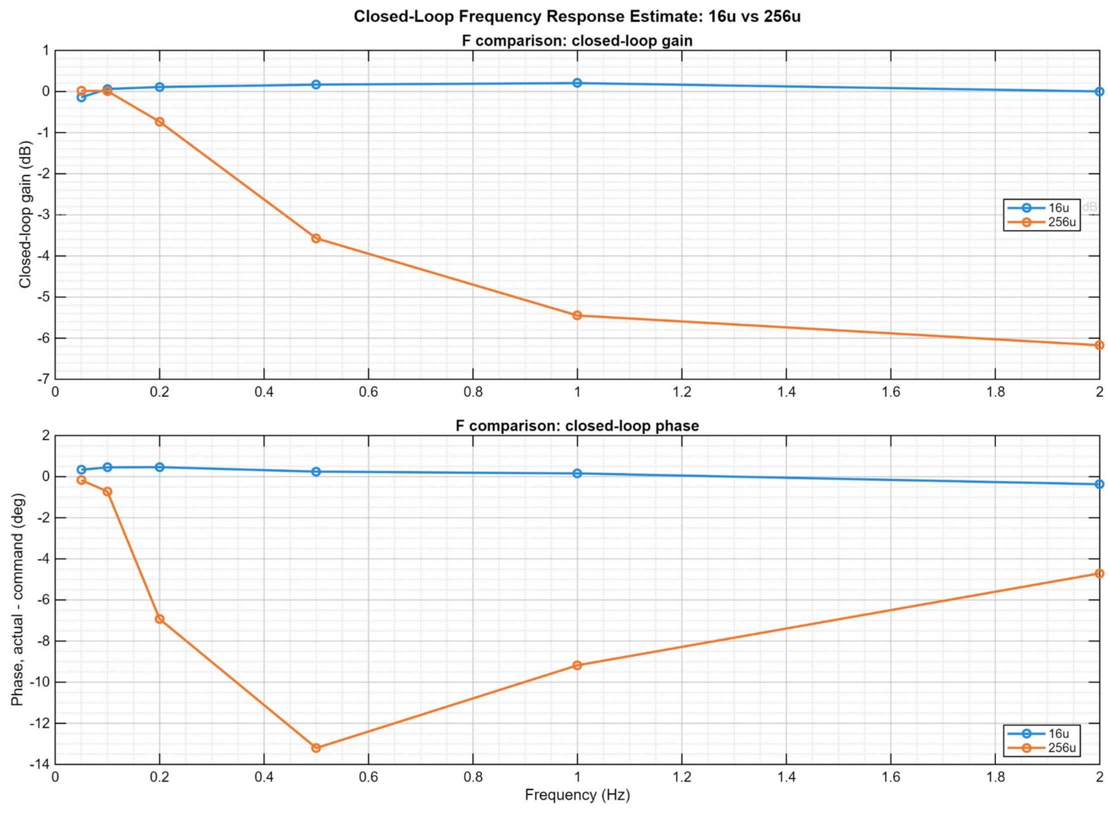

Figure 12. Closed-loop actual/command response over approximately 0.33–2 Hz.

These results show the firmware, physical stage, timestamped acquisition, host logging, and analysis pipeline operating as one system. The frequency result is a closed-loop characterization, not a claim of open-loop plant identification or formal phase margin.

11. Delivered Capabilities and Release Actions

Layer	Delivered capability
Dual-core firmware	OpenAMP/RPMsg coordination, shared-memory ring, HSEM startup, coherency policy, sequence/drop/fault accounting
Motion firmware	Pulse generation, encoder capture, correction, automated sequencing, heartbeat, deadman, and overrun monitoring
Host application	Connection, commands, live plotting, orchestration, logging, metadata, and recovery
Analysis	Low-speed, tracking, step response, Allan deviation, and closed-loop sine-response processing
Release item	**Required action**
Ring allocation	Reduce capacity to 4,095 or reserve more than 256 KiB.
Integrated build	Tag and preserve one release binary/build proving final module integration.
IPC timing	Instrument end-to-end latency and worst-case jitter.
External telemetry	Use binary framing or faster transport for sustained 100 Hz full records.
Power and EMC	Measure idle/peak power and complete EMC/environmental validation.

12. Attribution and Revision Control

Herder Elektronische Systemen developed the Dual-Core Optical Motion Testbed dual-core firmware architecture, motion-control integration, host software, logging and analysis workflows, and customer-facing documentation represented here. No third-party affiliation or endorsement is claimed.

Rev	Date	Description
2.9	2026-08-04	Renamed the public project, removed the retired institutional identifier, sanitized GUI evidence, and preserved the verified dual-core firmware, host-software, and results content.